

Biconnected Graph Triangulation Algorithms

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Complete Algorithms

FindAllTriangulations

Algorithm 1 FindAllTriangulations

```
1: procedure FINDALLTRIANGULATIONS( $G, k$ )
2:   Label faces as  $F_1, F_2, \dots, F_k$  arbitrarily
3:   Initialize global chord set  $S$ 
4:    $T \leftarrow \emptyset$  ▷ Empty partial triangulation
5:   FINDALLTRIANGULATIONS_CYCLE( $F, T, 0$ )
6: end procedure
```

FindSafeRootTriangulation

Algorithm 2 FindSafeRootTriangulation

```
1: procedure FINDSAFEROOTTRIANGULATION( $F, \text{index}$ )
2:   startIndex  $\leftarrow n - 1$ 
3:   endIndex  $\leftarrow 0$ 
4:   while startIndex > endIndex + 1 do
5:     if chord ( $v_{\text{startIndex}}, v_{\text{endIndex}}$ ) exists in  $S$  then
6:       startIndex  $\leftarrow \text{startIndex} - 1$ 
7:     else
8:       endIndex  $\leftarrow \text{endIndex} + 1$ 
9:     end if
10:  end while
11:   $v_r \leftarrow v_{\text{endIndex}}$  ▷ Root vertex for safe triangulation
12:  Build fan triangulation  $T_r$  from  $v_r$  as root vertex
13:  return  $T_r$ 
14: end procedure
```

Time Complexity: $\mathcal{O}(n)$ where n is the number of vertices in the face.

FindAllTriangulationsCycle

Algorithm 3 FindAllTriangulationsCycle

```
1: procedure FINDALLTRIANGULATIONS-cycle( $F, T, i$ )
2:    $T_i \leftarrow \text{FINDSAFEROOTTRIANGULATION}(F, i)$   $\triangleright \mathcal{O}(n)$ 
3:   if  $F_i$  is the last face then  $\triangleright$  All faces triangulated
4:     output ( $T \cup T_i$ )
5:   else
6:     FINDALLTRIANGULATIONS-cycle( $F, T \cup T_i, i + 1$ )
7:   end if
8:   for  $j = n - 1$  down to 3 do
9:     FINDALLCHILDTRIANGULATIONS-cycle( $T_i(v_r, v_j), T, i$ )
10:  end for
11: end procedure
```

FindAllChildTriangulationsCycle

Algorithm 4 FindAllChildTriangulationsCycle

```
1: procedure FINDALLCHILDTRIANGULATIONS-cycle( $T_i, T, i$ )
2:   if  $F_i$  is the last face then  $\triangleright$  All faces triangulated
3:     output ( $T \cup T_i$ )
4:   else
5:     FINDALLTRIANGULATIONS-cycle( $F, T \cup T_i, i + 1$ )
6:   end if
7:   Let  $(v_b, v_{b'})$  be the leftmost generating chord of  $T_i$ 
8:   for  $j$  from  $b$  to  $n - 1$  do
9:     if  $(v_r, v_j)$  is a chord of  $T_i$  and  $\text{oppositePair}(v_r, v_j) \notin S$  then
10:      FINDALLCHILDTRIANGULATIONS-cycle( $T_i(v_r, v_j), T, i$ )
11:    end if
12:  end for
13: end procedure
```
